Practice workbook · Explain antigen-antibody specificity and how monoclonal antibodies are made

What antigens are, why each antibody binds one antigen, reactions to self or foreign proteins, and producing monoclonal antibodies from fused cells.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIGEN-ANTIBODY-SPECIFICITY

1. Question 4 · part b · 3 marks

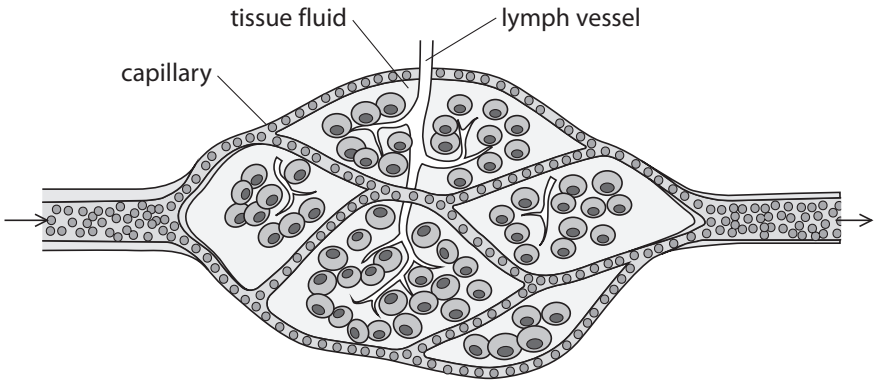
January 2023 Paper 2

Worked-example retry · Attempt part b only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

(2)

1 .....

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2 .....

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(ii) Describe how tissue fluid is formed.

(4)

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1. Question 4 · part b · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part b only.

(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

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(Total for Question 4 = 11 marks)

1. Question 4 · part b · 3 marks · Working and answer

Working and answer

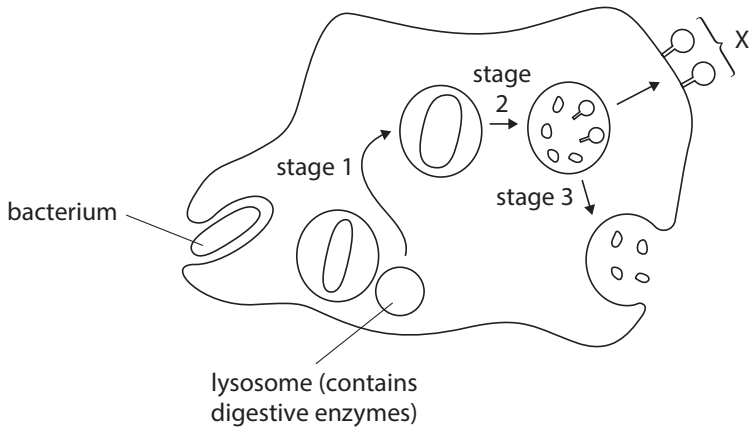
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2. Question 6 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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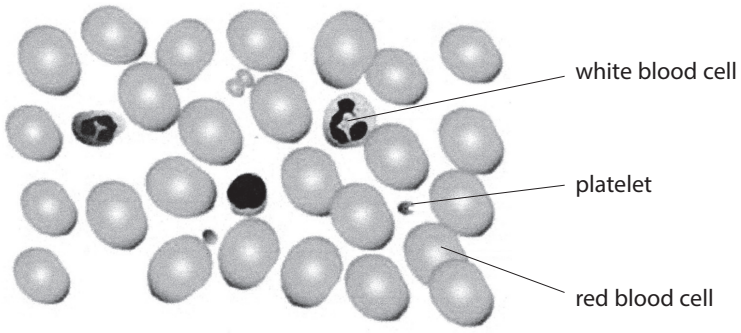
2. Question 6 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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2. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

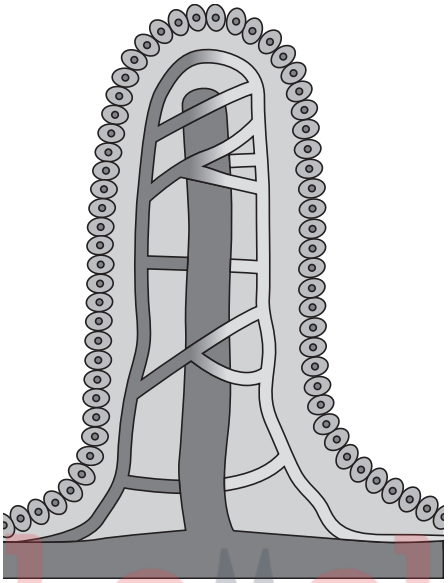
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3. Question 7 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

7 The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

(3)

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3. Question 7 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

Describe how the immune system of people with coeliac disease damages the lining of the small intestine.

(2)

(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

(3)

(Total for Question 7 = 8 marks)

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3. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 7 · part d · 2 marks

June 2025 Paper 2R

Further practice · Attempt part d only.

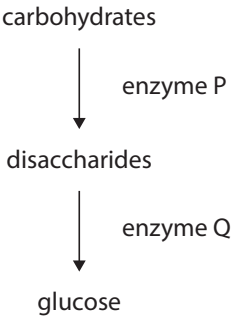
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

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(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

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(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

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4. Question 7 · part d · 2 marks

June 2025 Paper 2R

Further practice · Attempt part d only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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Working and answer

5. Question 8 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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5. Question 8 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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5. Question 8 · part a.iv · 1 mark · Working and answer

Working and answer

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6. Question 8 · part a.v · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.v only.

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(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

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(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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6. Question 8 · part a.v · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.v only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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6. Question 8 · part a.v · 2 marks · Working and answer

Working and answer

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7. Question 8 · part whole · 5 marks

January 2020 Paper 2

Further practice · Attempt part whole only.

8 Describe how monoclonal antibodies are produced.

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(Total for Question 8 = 5 marks)

7. Question 8 · part whole · 5 marks · Working and answer

Working and answer

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